

BIOL 343

Applied Bioinformatics

Python environments

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Managing projects with environments

Problems when doing bioinformatics:

1. Operating system pollution

*Operating system allows multiple versions of language interpreters/
code libraries to be installed at the same time*

2. Discrepant dependencies

Not all language interpreters/libraries work on all computers

3. Conflicting packages

Not all libraries work with each other

4. Irreproducibility

*Analyses performed on your computer should be able to be easily
reproduced on a different computer*



Virtual environments

Managing Python projects with environments

```
$ which python  
/data/software/python-libs/  
3.0.0/bin/python  
$ python --version
```

```
$ module load python-libs  
$ source activate ds250  
$ which python  
~/.conda/envs/DEEPLABCUT/bin/python  
$ python --version  
Python 3.10.14  
$ python -c 'import numpy; print(numpy.version.version)'  
1.26.4
```

```
$ conda activate biol343  
$ which python  
~/.conda/envs/biol343/bin/python  
$ python --version  
Python 3.12.3  
$ python3 -c 'import numpy; print(numpy.version.version)'  
1.26.4
```

Managing Python projects with environments

Many package/environment managers:

1. Native virtual environments (venv)
2. Virtualenv
3. Anaconda/Miniconda/Mamba/Micromamba

Advantages to the conda family:

1. Not just for Python (can manage R interpreters, R libraries, and other command line tools)
2. Very popular within the bioinformatics community (i.e., the bioconda channel)
3. Strong support base
4. Mamba is installed on BOSE

Conda envs in BIOL 343

- Each student will use their own conda env
- Start with the **biol343** foundation (find in the GitHub repo)
- Can install your own packages

Live code activity...

1. Create new biol343 env
 1. `mamba env create -f env.yml`
2. Activate env
 1. `source activate biol343`
3. Add bioconda and conda-forge channels
 1. `conda config --add channels bioconda`
4. Install a new package (seqkit)
 1. `mamba install seqkit`
5. Test the new package
 1. `seqkit --help`
6. Demonstrate that the default env didn't change
 1. `conda deactivate`
 2. `seqkit --help`